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Research on Credit Risk Evaluation of Commercial Banks Based on Artificial Neural Network Model

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Abstract

Credit risk, as the main risk currently faced by commercial banks, is largely determined by whether the borrower can repay the credit loan on schedule. Therefore, the research on credit risk of commercial banks has important theoretical and practical significance. This article mainly predicts the credit risk of corporate customers of commercial banks by constructing Artificial Neural Network model. First, this article selects 14 financial indicators to construct a credit risk evaluation index system based on the results of previous studies; second, combined with cluster analysis and factor analysis to determine the actual credit rating of the sample data, thereby giving the sample data the category label; finally, combining the above research, established and compared two traditional credit risk prediction models and three commonly used Artificial Neural Network models, and finally compared the prediction performance to select the best model to achieve prediction and evaluation of the credit risk of corporate customers of commercial banks.

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Keywords: Cluster analysis,Factor analysis, Artificial Neural Network,Commercial banks,Credit risk.

1. Introduction

The main risks currently faced by commercial banks are credit risk, interest rate risk and operational risk. Credit risk is the most important aspect. It mainly refers to the risk that borrowers cannot perform their obligations in accordance with the contract and cannot repay bank loans on time or in full. Generally speaking, the main reasons for credit risk are the external macro factors and the specific internal operating conditions of the company. The specific situation of the company in these two aspects will obviously be listed in the company's financial statements as financial indicators.

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Regarding how to evaluate the credit risk of a company through its financial indicators, previous researchers have made various attempts. Wang et al. analyzed the existing problems of current credit evaluation indicators, and proposed several ideas for perfecting current financial indicator system[1]. In recent years, the Artificial Neural Network model has attracted more and more researchers' attention due to its advantages of non-linearity, parallel computing, high fault tolerance, and good generalization performance. Shan took the banking institutions of Shandong Province as an example and established a BP Neural Network model to evaluate the risk of green credit loans[2]. Li screened the characteristics of risk indicators and built a deep fully connected Neural Network to give early warning of extreme risks in the stock market[3]. Since the training of Neural Network models requires category labels, it is reflected in the credit risk assessment that the actual credit rating of the target company needs to be determined first, but previous studies are lacking in this respect.

2. Credit risk indicator system

2.1. Selection of financial indicators

Corporate financial statements contain data on various financial indicators of the company during a period of operation. However, only some of indicators can reflect the credit risk of the enterprise. So this article follows certain indicator selection principles.

2.1.1. Index selection principle

- Comprehensiveness: There are many factors that affect corporate credit risk. When selecting indicators, it must be ensured that the selected indicators include most of the key influencing factors.
- Industry adaptability: There are huge differences between companies of different types and industries. Therefore, it is necessary to select the applicable financial indicators for different types of industries.
- Importance: Under the premise of ensuring comprehensiveness, focuses on selecting financial indicators that have a high impact on credit risk for modeling and evaluation.

2.1.2. Index selection.

This article selects the following financial indicators to construct an evaluation indicator system: current ratio, quick ratio, debt asset ratio, equity ratio, return on net assets, operating profit ratio, price to earnings ratio, total asset turnover, accounts receivable turnover, inventory turnover, current assets turnover, operating income growth rate , total assets growth rate, operating profit growth rate.

2.2. Data collection and preprocessing

The data in this article comes from the CSMAR database,selecting the top 150 listed companies in the manufacturing industry and eliminate abnormal companies such as missing annual reports. There are 133 data items remaining, each with 14 indicators. Finally, use X1-X14 to label the indicators.

Next, the remaining 133 pieces of data are preprocessed. Because some of the data have missing values, this article uses the sequence average to replace the missing values.

3. Actual credit rating evaluation based on cluster analysis and factor analysis

Since training the Artificial Neural Network classifier requires the category label of the sample data. Therefore, a key issue is how to determine the actual credit rating of each company during the year.

In view of this problem, many researchers put forward their own solutions. Jing used the 3δ rule to determine the credit risk status. By calculating the mean μ and standard deviation δ of enterprise factor scores, the credit

risk status was determined into five grades with μ+δ, μ-δ, μ-3δ and μ-3δ as the criticality[4]. Yi used the entropy method to calculate the actual credit risk value of the enterprise, and judged the entropy value at (0,0.5),(0.5,0.75) and (0.75,1) as the three levels of low risk, medium risk and high risk respectively. Finally, the predicted value was compared with the actual value, and the prediction was considered accurate if the deviation was less than 10% [5]. Long chose the s-type function as the excitation function. because the value range of the s-type function is (0,1), he directly judged that the output is located at five levels of (0.8,1), (0.6,0.8), (0.4,0.6), (0.2,0.4) and (0,0.2), respectively, with the risk degree decreasing in turn[6].

This paper proposes a new solution to this problem: this paper adopts the method of cluster analysis to cluster 133 enterprises into five categories. Then, the dimension is reduced by factor analysis method and the total factor score of each enterprise is obtained. Since we can think that the credit risk of enterprises is normally distributed . Therefore, the actual credit risk grade of each category can be determined according to the degree to which the total score mean of each category of factors deviates from the total score mean of the whole factor. The degree of deviation increases successively from small to large credit risks, corresponding to AA, A, BB, B and C.

3.1. Cluster analysis

As two kinds of commonly used clustering methods, system clustering method and K-means clustering method have different application scenarios. In the case of large amount of data, the accuracy and processing efficiency of K-means clustering are higher than that of system clustering. In this paper, the number of clusters needed could be determined as 5 in advance, and the data processed at the same time were 133 pieces of 14-dimensional data, with a large amount of data, so the K-means clustering method was more appropriate.

Set the number of clusters to 5, and set the maximum number of iterations to 10 for clustering. Convergence is reached in the 4th iteration, the cluster center does not change, and the clustering ends. The final number of samples in each category is shown in the figure below:

Table 1. The result of clustering

Category name	1	2	3	4	5
quantity	7	2	121	1	2

3.2. Factor analysis

3.2.1. KMO and Bartlett tests.

Before the factor analysis, KMO and Bartlett tests should be performed on the data to test the reliability and validity of the data and judge whether the data can be used for factor analysis. The test results of the data in this paper are shown in the table below. After comparison, the data in this paper have passed the test.

Table 2. The result of KMO and Bartlett tests

Kmo test coefficient		.562
Bartlett test	Approximate chi square	1426.455
	Free degree	91
	Significance	.000

3.2.2. Factor analysis.

Principal component method was used to extract initial common factors, maximum variance method was used to perform factor rotation, and finally regression method was used to obtain the calculation equation of factor score. The final interpretation result of the total variance is shown in the figure below:

	Initial eigenvalue			Extraction			Rotation		
	Total	Variance%	Accumulate %	Total	Variance%	Accumulate %	Total	Variance%	Accumulate %
1	3.389	24.210	24.210	3.389	24.210	24.210	2.590	18.499	18.499
2	2.652	18.944	43.154	2.652	18.944	43.154	2.510	17.927	36.426
3	1.884	13.455	56.609	1.884	13.455	56.609	2.325	16.610	53.036
4	1.523	10.882	67.491	1.523	10.882	67.491	2.020	14.432	67.468
5	1.115	7.963	75.455	1.115	7.963	75.455	1.118	7.987	75.455
6	.941	6.722	82.177						
7	.774	5.531	87.707						
8	.574	4.102	91.809						
9	.415	2.966	94.775						
10	.346	2.473	97.248						
11	.237	1.692	98.940						
12	.111	.795	99.736						
13	.027	.190	99.925						
14	.010	.075	100.000						

Fig. 1. Total Variance Explained

When the cumulative variance contribution rate reaches 75%, the common factors are considered to be representative and retain most of the information of the original variables. Therefore, it is appropriate to choose the first five components as the final common factors in this paper.

Finally, this paper also needs to calculate the total factor score of each enterprise. The specific calculation method is to multiply the variance contribution rate of each common factor by the factor score of each enterprise on each common factor. If F represents the total factor score, and f1 to f5 represent the factor scores of each common factor respectively, then the formula for calculating the total factor score is as follows:

$$F = 0.18499 * f1 + 0.17927 * f2 + 0.16610 * f3 + 0.14432 * f4 + 0.07987 * f5 \quad (1)$$

3.2.3 Actual credit rating

After the F value of all enterprises is calculated, the average value of the total score of all enterprises' factors is -.0000. Finally the actual credit rating are as shown in the figure below:

Table 3. Actual credit rating

Category name	Quantity	Mean	Actual credit rating
1	7	.2739	A
2	2	.3849	BB
3	121	-.0405	AA
4	1	1.3000	C
5	2	.4623	B

4. Traditional credit risk forecasting model

In this paper, commonly used traditional credit risk prediction models are tested to verify their prediction accuracy, and the prediction effect is compared with the Artificial Neural Network model constructed below.

4.1. Decision tree model

Decision tree CHAID algorithm is used in this paper.The data set was randomly divided into training set and test set according to 7:3. Numbers 1-5 were used to represent the five credit risk levels of AA, A, BB, B and C, and the final classification prediction results were shown as follows.It can be seen that the decision tree model has a low prediction accuracy for the data in this paper and almost no ability to predict abnormal credit risks.

Sample	Real	Prediction					Accuracy
Train	1	83	0	0	0	0	100.0%
	2	6	0	0	0	0	0.0%
	3	1	0	0	0	0	0.0%
	4	2	0	0	0	0	0.0%
	5	1	0	0	0	0	0.0%
	Percentage	100.0%	0.0%	0.0%	0.0%	0.0%	89.2%
Text	1	38	0	0	0	0	100.0%
	2	1	0	0	0	0	0.0%
	3	1	0	0	0	0	0.0%
	4	0	0	0	0	0	.
	5	0	0	0	0	0	.
	Percentage	100.0%	0.0%	0.0%	0.0%	0.0%	95.0%

Fig. 2. The result of Decision tree

4.2. Discriminant analysis model

This paper constructs a discriminant analysis model with all sample data.Numbers 1-5 represent the five credit risk grades of AA, A, BB, B and C, and the final classification prediction results are shown as follows. It can be seen that the discriminant analysis model has better prediction performance for the data in this paper compared with the decision tree model.

Sample	Prediction	1	2	3	4	5	Accuracy
	1	119	2	0	0	0	98.4%
	2	3	4	0	0	0	57.1%
	3	1	0	1	0	0	50.0%
	4	1	0	0	1	0	50.0%
	5	1	0	0	0	0	0.0%
							93.9%

Fig. 3. The result of Discriminant analysis

5. Artificial Neural Network model

For credit risk evaluation research, predecessors have tried many methods. Li used AHP to evaluate the credit risk of small and medium-sized enterprises under the supply chain model[8]. Ju et al. demonstrated the feasibility of expert judgment scoring method in bank credit granting business from an empirical perspective[9].Wang et al. evaluated the credit risk of small and medium-sized enterprises by using the multi-level fuzzy comprehensive evaluation method[10]. With the continuous development of artificial intelligence, machine learning theory and technology. More and more researchers use nonlinear neural network model to evaluate credit risk. Alapedes et al. took advantage of the neural network model in processing nonlinear signals and applied it to the credit risk management of commercial banks for the first time[11]. Yang et al. first proposed a bank loan risk management system based on neural network and expert system[12].Han et al. took financial and non-financial factors into consideration and used fuzzy neural network to forecast credit risk portfolio[13].Since then, a large number of researchers began to use neural network model to study credit risk assessment.

This paper tests three kinds of Artificial Neural Networks one by one, and finally selects the model most suitable for the data in this paper. Python is used to build the BP network, and the perceptron network and radial basis function network are completed by SPSS which saves unnecessary repetitive work.

5.1. BP neural network model

5.1.1 Topology

For the problem to be dealt with in this paper, a single hidden layer neural network is sufficient, and on the determination of the number of nodes in each layer: firstly, the number of nodes in the input layer of the model is set as 5 for 5 factors; Secondly, the number of nodes in the output layer can be set as 5 for 5 credit risk levels, the risk levels AA, A, BB, B and C can be represented by (1,0,0,0,0), (0,1,0,0,0), (0,0,1,0,0), (0,0,0,1,0) , (0,0,0,0,1). The number of nodes in the hidden layer was determined as 9 after several attempts.

5.1.2. Parameters and related functions

In this paper, the learning rate is set to 0.4 and the maximum number of iterations is set to 10000.And the relevant functions in this article's Python code are set as follows:

- Initialization parameters: Random. randn is used to randomly generate a group of initial weight parameters following normal distribution, and the parameters are saved in a parameters dictionary
```
def initialize_parameters(n_x, n_h, n_y):
```
- Forward propagation: the activation function needs to be determined here. In this paper, the first layer of the activation function is selected as the hyperbolic tangent function; the second layer is selected as the Sigmoid function to get the output value between (0,1).
```
def forward_propagation(X, parameters):
```
- Calculation of cost function: The cross-entropy function is selected as the cost function in this paper.Using the cross entropy function as the cost function can improve the convergence speed when using the Sigmoid function as the activation function.
```
def compute_cost(a2, Y, parameters):
```
- Back propagation: Use the chain derivative rule to realize back propagation of error. The response errors of the hidden layer and the output layer are obtained by differentiating the activation response with the target output corresponding to the training input.
```
def backward_propagation(parameters, cache, X, Y):
```
- Update parameters: update the values of the neural network parameters, including the weight and bias of each layer, to realize the iteration of parameters.
```
def update_parameters(parameters, grads, learning_rate=0.4):
```
- Model evaluation: Compare and print out the actual results of the test set with the predicted results, return the location of the predicted error data and calculate the final prediction accuracy.
```
def predict(parameters, x_test, y_test):
```
- Build the neural network: build and train the neural network, and set the number of iterations as 10000.The cost function value is output every 1000 iterations.
```
def nn_model(X, Y, n_h, n_input, n_output, num_iterations=10000, print_cost=False):
```

5.1.3. Result

When the number of iterations reaches 9000, the value of the cost function is already very small, and the parameters can be considered to be close to the optimal at this time. The final prediction result is shown in the figure below. The overall prediction accuracy of the model is fair, reaching 92.5%, but it has a certain but not high accuracy in the prediction of abnormal loans.

```
Prediction
[[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
  1 1 1 1]
 [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
  0 0 0 0]
 [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
  0 0 0 0]
 [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
  0 0 0 0]
 [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
  0 0 0 0]]
 Real
[[1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 0 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0
  1 1 1 1]
 [0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
  0 0 0 0]
 [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
  0 0 0 0]
 [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
  0 0 0 0]
 [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
  0 0 0 0]]
12
14
35
Accuracy 92.50%
```

Fig. 4. The result of BP-ANN

5.2. Radial basis function network

5.2.1. Parameter selection

Before model training, various parameters are set as follows: the three-layer neural network with a topology structure of 5*5*5; The activation function of the hidden layer is set as Softmax function, and the activation function of the output layer is set as identity. Each layer is not set bias; The cost function is the sum of squares function. The partition method of training set and test set is 7:3 random partition.

5.2.2. Results

The final prediction results are shown in the figure below. The overall prediction accuracy of the model is good, with the prediction accuracy of the training set and the test set reaching 91% and 97.7%. Compared with BP Neural Network, the prediction accuracy of non-performing loans has been significantly improved.

		Prediction					
Sample	Real	(0,0,0,0,1)	(0,0,0,1,0)	(0,0,1,0,0)	(0,1,0,0,0)	(1,0,0,0,0)	Accuracy
Train	(0,0,0,0,1)	0	0	0	0	1	0.0%
	(0,0,0,1,0)	0	2	0	0	0	100.0%
	(0,0,1,0,0)	0	0	0	0	2	0.0%
	(0,1,0,0,0)	0	0	0	2	4	33.3%
	(1,0,0,0,0)	0	0	0	1	77	98.7%
	Percentage	0.0%	2.2%	0.0%	3.4%	94.4%	91.0%
Text	(0,0,0,0,1)	0	0	0	0	0	0.0%
	(0,0,0,1,0)	0	0	0	0	0	0.0%
	(0,0,1,0,0)	0	0	0	0	0	0.0%
	(0,1,0,0,0)	0	0	0	0	1	0.0%
	(1,0,0,0,0)	0	0	0	0	43	100.0%
	Percentage	0.0%	0.0%	0.0%	0.0%	100.0%	97.7%

Fig. 5. The result of Radial basis function network

5.3. Perceptron network

5.3.1. Parameter selection

Before the model training, the parameters are set as follows: the three-layer neural network with a topology structure of 5*11*4;The activation functions of hidden layer and output layer are set as Sigmoid function. Set bias in both input layer and hidden layer; The cost function is the sum of squares function. The initial learning rate is set at 0.01. The dividing method of training set and test set is 2:1 randomly.

One problem needs to be noted here: since there is only one enterprise with credit risk grade C in the data set, it is divided into the test set here and does not appear in the training set, so it is excluded in the training process, and the final neural network only has four outputs: AA, A, BB and B.

5.3.2. Results

The final prediction results are shown in the figure below. The overall prediction accuracy of the model is very good, and the prediction accuracy of the training set and the test set is up to 98.8% and 95.8%. It is particularly noteworthy that, compared with BP Neural Network and radial basis function network, the prediction accuracy of non-performing loans has been significantly improved. The prediction accuracy of A and BB loans has reached 100%, showing good prediction performance.

Sample	Real	Prediction				
		(0,0,0,1,0)	(0,0,1,0,0)	(0,1,0,0,0)	(1,0,0,0,0)	Accuracy
Train	(0,0,0,1,0)	0	0	0	1	0.0%
	(0,0,1,0,0)	0	1	0	0	100.0%
	(0,1,0,0,0)	0	0	3	0	100.0%
	(1,0,0,0,0)	0	0	0	79	100.0%
	Percentage	0.0%	1.2%	3.6%	95.2%	98.8%
Text	(0,0,0,1,0)	0	0	0	1	0.0%
	(0,0,1,0,0)	0	1	0	0	100.0%
	(0,1,0,0,0)	0	0	4	0	100.0%
	(1,0,0,0,0)	0	0	1	41	97.6%
	Percentage	0.0%	2.1%	10.4%	87.5%	95.8%

Fig. 6. The result of Perceptron network

Finally, by comparing the prediction effects of the three kinds of neural network models, it can be judged that commercial banks can use the perceptron neural network model to make risk prediction for the credit risk assessment of listed manufacturing enterprises, and the model can achieve a good prediction effect.

6. Conclusion

This paper combines cluster analysis and Artificial Neural Network model to study the credit risk evaluation of commercial banks, and compares the prediction performance of two traditional credit risk prediction models and three commonly used neural network models. Finally, it is found that the overall prediction performance of the Artificial Neural Network model is better for the problems in this paper, and the 5*11*4 three-layer perceptron neural network with a single hidden layer has the best prediction performance

References

[1] QX Wang, XW Wang. Improving financial index system to control bank credit risk [J]. Journal of North China Institute of Water Conservancy and Electric Power (Social Science Edition),2000(04):38-40.

[2] R Shan. Risk Assessment of Green Credit Based on BP Neural Network -- A Case Study of Shandong Banking Institutions [J].Modern Marketing (Information Edition),2020(06):102-103.

[3] QH Li. Empirical Research on Extreme Risk Warning Model of Stock Market Based on Artificial Intelligence Method [D].Capital University of Economics and Business,2019.

[4] ZR Jing. Research on Credit Risk Evaluation of Commercial Banks Based on BP Neural Network [D]. Shanxi University of Finance and Economics,2015.

[5] H Yi. Research on Agricultural Credit Risk Assessment of Commercial Banks Based on Neural Network [D]. Hunan University,2012.

[6] ZD Long. Research on Credit Risk Assessment of Commercial Banks Based on BP Neural Network [D]. Hubei University of Technology,2018.
[7] HQ He. Modern Statistical Analysis Method and Application (Fourth Edition)[M]. Beijing: China Renmin University Press,2016:201.
[8] B Li. Credit Risk Evaluation of Supply Chain Financing Mode Based on Analytic Hierarchy Process [D]. Tianjin University,2009.
[9] X Ju, SS Li. On the application of expert judgment and analysis method in bank credit granting business [J]. China Economist , 2008(02):206-207.
[10] S Wang, PT Yang, QW Huang. Credit risk assessment of small and medium-sized enterprises based on multi-level fuzzy comprehensive evaluation [J]. The Theory and Practice of Finance and Economics,2014,35(05):13-17.
[11] Lapedes.A, Farber.R. Nonlinear singal processing using neural networks: perdiction and System Modeling[J].Technical Report,Los Alamos National Laboratory,Los Alamos,New Mexico,1987.
[12] BA Yang, M Zhu. Research on the Decision of Bank Loan Risk Management Combined with Neural Network and Expert System -- National Natural Science Foundation Project 79770086 Recharge [J]. Chinese Journal of Management,2006(04):387-390.
[13] P Han,YM Zhu. Fuzzy neural network based credit risk portfolio forecasting [J]. The Journal of Quantitative & Technical Economics,2001(05):107-110.